

Decision Control Structures in C (if-else, switch, ternary)

[DEF] Definition to Remember

Decision control structures direct program execution along different execution paths based on the boolean evaluation of conditional expressions.

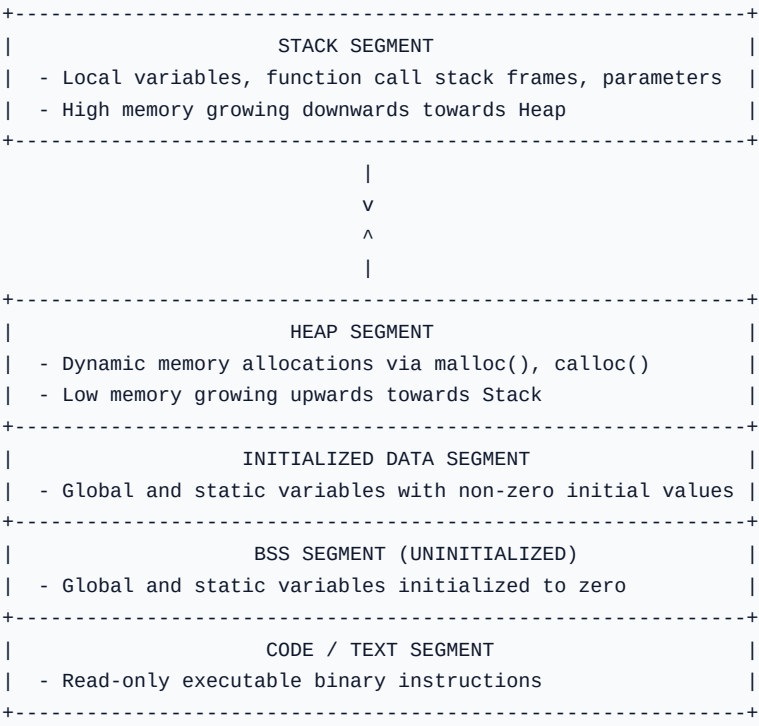
1. Concept Overview & Fundamentals

Decision Control Structures in C (if-else, switch, ternary) forms a core cornerstone of computer systems programming, hardware-level software engineering, and competitive problem solving.

In the C execution model:

2. Technical Architecture & Memory Mechanics

Memory Layout & Storage Organization



Core Architecture Breakdown

Component / Concept	Technical Specification	Operational Characteristics	Performance / Complexity
Storage Allocation	Stack vs Heap vs Data Segment	Stack is ultra-fast LIFO; Heap offers	Stack: O(1) push/pop; Heap: O(1)
Type Checking	Static Typing	Compiler verifies types during compila	Zero runtime type penalty
Addressing Model	Byte-addressable physical RAM	Pointer stores hexadecimal byte memor	Direct memory dereference (O(1))
Execution Flow	Procedural / Sequential	Controlled via jumps, branches, loops,	Deterministic instruction cycles

3. Core Syntax & Implementation Patterns

Essential Idioms & Header Declarations

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <stdbool.h>

// Core standard implementation pattern
int main(void) {
    // Explicit initialization
    int status = 0;

    // Core logic execution
    printf("Executing C Module: Decision Control Structure (if-else, switch, goto, ternary)\n");

    return status;
}
```

4. Real-World Applications & System Engineering

- * Operating System Kernels: Linux, UNIX, and Windows NT kernels utilize C for drivers, process scheduling, and memory paging.

* En

5. Comprehensive Solved Problems & Code Walkthroughs

Problem 1: AbsoluteTO

Problem Description & Objective:

C Code Implementation:

Create

//Crea

#inclu

int ma

/*

*

*

*

*

*

*

in

pr

sc

ab

pr

```
re  
}
```

Problem 2: CategorizeAgeGroup

Problem Description & Objective:

Implem

C Code Implementation:

```
/*  
Create  
  
Child  
Teen -  
Adult  
Senior  
*/  
#inclu  
  
int ma  
/*  
*  
*  
*  
Adult  
*  
*  
*  
*  
in  
  
pr  
sc  
  
if  
  
}  
  
}  
  
}  
  
}  
re  
}
```

Problem 3: GradeCalculator

Problem Description & Objective:

Implem

C Code Implementation:

```

/*
Create
A -> a
B -> a
C -> a
D -> a
F -> b
*/

#include
int main
/*
*
*
*
*
-> about
*
*
*
*

fl

pr
so

if

}

}

}

}

}

}

re
}

```

Problem 4: GreatestNumber

Problem Description & Objective:

C Code Implementation:

Create

```
//Crea
#include
int ma
```

```
/*
 *
 *
 *
 *
 *
 *
 */
in
pr
sc
pr
sc
pr
sc
if
}e
}e
}
re
}
```

Problem 5: LeapYear

Problem Description & Objective:

Create
400).

C Code Implementation:

```
//Crea
100, u

#inclu

int ma
/*
 *
 *
by 4 b
 *
 *
 *
 *

in
pr
sc

if

}
```

```
}  
re  
}
```

Problem 6: MinimumNumberTO

Problem Description & Objective:

C Code Implementation:

Create

```
//Crea  
  
#inclu  
int ma  
/*  
*  
*  
*  
*  
*  
*  
*  
  
in  
  
pr  
sc  
pr  
sc  
  
(n  
  
re  
}
```

Problem 7: NumberChecker

Problem Description & Objective:

C Code Implementation:

Create

```
//Crea  
  
#inclu  
int ma  
/*  
*  
*  
*  
*  
*  
*  
*  
*  
  
in  
  
pr
```

```
sc
if
}e
}e
}
re
}
```

Problem 8: OddEven

Problem Description & Objective:

Create

C Code Implementation:

```
//Crea
#include
int ma
/*
*
*
*
*
*
*
*
in
pr
sc
if
}e
}e
}
re
}
```

Problem 9: OddEvenTO

Problem Description & Objective:

Create

C Code Implementation:

```
//Crea
#include
int ma
/*
*
*
```

```
*
*
*
*
*
*
in
pr
sc
(n
re
}
```

Problem 10: PrintMonthSwitch

Problem Description & Objective:

C Code Implementation:

Create

```
//Creat
#include
int ma
/*
*
*
*
*
*
*
in
pr
sc
Sw
```

```
}  
re  
}
```

Problem 11: ScoreCategoryTO

Problem Description & Objective:

Create
> 80, M

C Code Implementation:

```
//Crea  
operat  
  
#inclu  
  
int ma  
/*  
*  
  
ternar  
*  
*  
*  
*  
  
in
```

```
pr
sc

(s

re

}
```

Problem 12: SimpleCalculatorSwitch

Problem Description & Objective:

C Code Implementation:

Create
multipl

```
//Crea
like a

#inclu

int ma
/*
*
*
operat
*
*
*
*

in
ch

pr
sc

pr
sc

pr
sc

sw
```

6. Exam Scoring Points & Memory Revision

[MUST-WRITE] Must-Write Points

1. if-else statements evaluate non-zero as true and zero as false in C.
2. switch statement evaluates integral/char expressions; each case requires break to prevent fall-through.
3. default case in switch handles unmatched values.
4. Ternary operator (:) provides an inline 3-operand conditional expression.
5. Avoid goto statements to prevent spaghetti code and maintain structured control flow.

[RECALL] Quick Recall

Boolean Test -> if / else if / else -> switch(integral) + break -> Fallback default